Septic shock in severe CAP and vasopressors — June Q1

Outline your resuscitation and antimicrobial strategy in the first hours

- Balanced crystalloid 30 mL/kg in the first 3 h, further fluid only if fluid-responsive (dynamic tests, CRT, lactate trend)
- Target MAP ≥ 65 mmHg; noradrenaline early, peripheral line acceptable until a central line is placed
- Cultures (blood × 2, sputum/ETA), urinary pneumococcal and Legionella antigens, respiratory PCR – without delaying antibiotics
- Antibiotics within 1 h: β-lactam (e.g., ceftriaxone 2 g) + macrolide (azithromycin 500 mg); antipseudomonal/anti-MRSA cover only if risk factors
- Source/organ support: lung-protective ventilation if intubated, lactate re-measured within 2–4 h
- Hydrocortisone 200 mg/day if noradrenaline ≥ 0.25 µg/kg/min for ≥ 4 h; in severe CAP early hydrocortisone reduces mortality (CAPE COD 2023)

Compare the mechanism of action of noradrenaline and vasopressin

- Noradrenaline: potent α1 agonist (Gq → IP3 → intracellular Ca2+ → vasoconstriction) with modest β1 inotropy
- Vasopressin: V1a receptor on vascular smooth muscle (Gq → IP3/Ca2+), independent of adrenergic receptors
- Septic shock causes relative vasopressin deficiency; vasopressin also closes K-ATP channels and restores catecholamine sensitivity
- Vasopressin may spare the kidney (efferent > afferent constriction); can cause digital/mesenteric ischaemia

Septic shock in severe CAP and vasopressors — June Q1 (cont.)

Compare the mechanism of action of adrenaline and angiotensin II

- Adrenaline: α1, β1 and β2 agonist → vasoconstriction plus inotropy/chronotropy
- β2 stimulation of muscle glycolysis raises lactate (type B), tachyarrhythmias and hyperglycaemia
- Angiotensin II: AT1 receptor (Gq → IP3/DAG) → arteriolar constriction, aldosterone and vasopressin release
- Angiotensin II raised MAP in catecholamine-refractory vasodilatory shock (ATHOS-3); thrombosis risk, avoid in mesenteric ischaemia

Describe your stepwise vasopressor escalation with supporting trial evidence

- Noradrenaline first line (SOAP II: dopamine more arrhythmias); MAP 65 is enough (SEPSISPAM; 65 trial in ≥ 65 years)
- Add vasopressin 0.03 U/min when noradrenaline 0.25–0.5 µg/kg/min (VASST, VANISH: noradrenaline-sparing, no mortality difference)
- Echo to look for septic cardiomyopathy: add dobutamine or switch/add adrenaline if low cardiac output
- Refractory shock: angiotensin II (ATHOS-3) or methylene blue as rescue; re-check source, fluids, adrenal function

Key point: Noradrenaline first; add vasopressin at 0.25–0.5 µg/kg/min; hydrocortisone if shock persists.

Upper GI bleeding on DAPT after STEMI — June Q2

Discuss resuscitation, transfusion targets and the evidence for haemostatic adjuncts (desmopressin, platelets, tranexamic acid)

- Airway protection if ongoing haematemesis or reduced consciousness; two large-bore IV lines, cross-match, activate massive haemorrhage protocol if needed
- Restrictive transfusion (Hb 7–8 g/dL) in GI bleeding generally; in recent ACS aim higher, around Hb 8–10 g/dL (MINT 2023 favoured liberal in MI)
- IV PPI bolus then infusion or high-dose intermittent; erythromycin 250 mg IV 30–120 min before endoscopy
- Platelet transfusion is not routinely recommended to reverse antiplatelet drugs in GI bleeding; ticagrelor's active drug also inhibits transfused platelets (PATCH showed harm in ICH, not GIB)
- Tranexamic acid not recommended: HALT-IT (12,009 pts) no mortality benefit and more venous thromboembolism
- Desmopressin has no good evidence in GI bleeding on antiplatelets; may be considered only as rescue
- Correct calcium and temperature; balanced product ratio if massive transfusion

Upper GI bleeding on DAPT after STEMI — June Q2 (cont.)

Discuss stent thrombosis risk and when to resume aspirin and the P2Y12 inhibitor

- Recent DES (< 30 days) with stopped DAPT has high stent thrombosis risk with high mortality (MI, death)
- Continue aspirin if possible; if stopped, resume as soon as haemostasis is secured, ideally within 24–72 h (stopping it raised death in Sung 2010)
- Hold ticagrelor only while bleeding is uncontrolled; resume within about 3–5 days of haemostasis, joint cardiology–GI decision
- Consider switching to clopidogrel (less potent) and keep long-term PPI co-therapy

Outline medical, endoscopic, radiological and surgical management

- Endoscopy within 24 h, earlier (< 12 h) after resuscitation if haemodynamically unstable; Forrest classification guides therapy
- Combined endoscopic therapy: adrenaline injection plus a mechanical (clips) or thermal method; haemostatic powder or over-the-scope clip as rescue
- Re-bleeding: repeat endoscopy once; then transcatheter arterial embolisation (first salvage)
- Surgery (under-running of the ulcer) if embolisation fails or is unavailable; post-endoscopy high-dose PPI for 72 h, test and treat H. pylori

Key point: No platelets, no TXA (HALT-IT), no good evidence for desmopressin; PATCH was in ICH, not GI bleeding. Restart aspirin early.

Refractory status epilepticus, PRIS, ketamine and pyridoxine — June Q3

Discuss the use, dosing and monitoring of midazolam, propofol and barbiturate infusions

- First line benzodiazepine (lorazepam 4 mg IV, midazolam 10 mg IM/IV, repeat once); second line levetiracetam 60 mg/kg, fosphenytoin 20 mg PE/kg or valproate 40 mg/kg (ESETT: equal efficacy)
- Refractory SE: intubate, start continuous EEG; target seizure control or burst suppression for 24–48 h then wean
- Midazolam 0.2 mg/kg bolus then 0.05–2 mg/kg/h: haemodynamically stable but tachyphylaxis and accumulation in renal failure
- Propofol 1–2 mg/kg then 2–5 mg/kg/h: fast on/off; hypotension, hypertriglyceridaemia, avoid > 4 mg/kg/h for > 48 h (PRIS)
- Thiopental/pentobarbital: very effective but hypotension, myocardial depression, ileus, immunosuppression and prolonged half-life
- Monitor arterial pressure, vasopressor need, lactate, CK, triglycerides, and EEG for non-convulsive seizures
- Continue maintenance antiseizure drugs and search for the cause in parallel

Refractory status epilepticus, PRIS, ketamine and pyridoxine — June Q3 (cont.)

Describe the pathophysiology, features and management of propofol infusion syndrome

- Propofol impairs mitochondrial fatty-acid oxidation and the respiratory chain → cellular energy failure in heart and muscle
- Risk: dose > 4 mg/kg/h for > 48 h, young age, catecholamines or steroids, low carbohydrate intake, critical illness
- Features: unexplained metabolic (lactic) acidosis, rhabdomyolysis, hyperkalaemia, hypertriglyceridaemia, AKI, hepatomegaly, Brugada-like ECG, bradyarrhythmia and heart failure
- Stop propofol immediately; supportive care, glucose supply, RRT for acidosis/hyperkalaemia, pacing or ECMO for cardiac failure (lipid emulsion is NOT a treatment)

Explain the rationale and indications for ketamine and pyridoxine

- Prolonged SE internalises GABA-A receptors and up-regulates NMDA receptors → benzodiazepine resistance
- Ketamine (NMDA antagonist) targets this mechanism; bolus 1–3 mg/kg then 1–10 mg/kg/h; supports blood pressure and allows lower vasopressor doses
- Pyridoxine (B6) is the cofactor for GAD, which makes GABA; isoniazid toxicity depletes it
- Give pyridoxine 1 g per g of isoniazid ingested (5 g IV if unknown); also in infant pyridoxine-dependent epilepsy

Key point: PRIS: stop propofol and support; lipid emulsion is NOT a treatment. Ketamine targets NMDA up-regulation.

CRRT versus intermittent haemodialysis and anticoagulation — June Q4

Compare CRRT and intermittent haemodialysis in outcome, haemodynamics and solute clearance

- Mortality: no difference between CRRT and IHD in RCTs and meta-analyses (e.g., Hemodiafe 2006)
- CRRT gives slower fluid and solute removal → better haemodynamic tolerance in shock; preferred in haemodynamic instability (KDIGO)
- CRRT avoids rapid osmolar shifts and cerebral oedema → preferred in brain injury, acute liver failure (raised ICP)
- IHD clears small solutes faster: better for urgent hyperkalaemia and dialysable poisoning, needs fewer nursing resources and allows mobilisation
- Dose: delivered effluent 20–25 mL/kg/h is enough (RENAL, ATN trials); prescribe ~25–30 to allow for downtime
- Observational data suggest more dialysis dependence after IHD as first modality, but not proven in RCTs
- Timing: no benefit of very early start (STARRT-AKI, AKIKI); harm with very late start (AKIKI-2)

CRRT versus intermittent haemodialysis and anticoagulation — June Q4 (cont.)

Compare regional citrate, heparin and no-anticoagulation strategies for CRRT

- Regional citrate: chelates ionised calcium in the circuit (post-filter iCa 0.25–0.35 mmol/L), calcium is replaced systemically (iCa 1.0–1.2)
- KDIGO recommends citrate first line when not contraindicated: longer filter life, less bleeding than heparin
- Citrate accumulation in liver failure or shock (poor metabolism): total/ionised Ca ratio > 2.5, rising anion gap and acidosis
- Other citrate risks: metabolic alkalosis (citrate → bicarbonate), hypo-/hypercalcaemia, hypernatraemia
- Unfractionated heparin: cheap and reversible, aPTT 1.5× or anti-Xa monitoring; risks bleeding and HIT
- Monitoring: post-filter and systemic iCa every 6 h, adjust citrate and calcium infusions; daily total Ca, Na, bicarbonate
- No anticoagulation: when bleeding or severe coagulopathy; use predilution, high blood flow, minimise circuit interruptions

Key point: Same mortality; CRRT for shock and raised ICP; citrate first line, watch total/ionised Ca > 2.5.

Weaning failure, diaphragm ultrasound, P0.1 and CROP — June Q5

List the causes of weaning failure by system

- Respiratory load: bronchospasm, secretions, high resistance (small ETT), low compliance (oedema, effusion, atelectasis), auto-PEEP
- Cardiac (weaning-induced pulmonary oedema): negative intrathoracic pressure raises venous return and LV afterload, plus catecholamine surge → ischaemia, raised LV filling pressure
- Neuromuscular: ICU-acquired weakness (CIP/CIM), diaphragm dysfunction (ventilator-induced, phrenic injury), residual sedation/NMB
- Metabolic: low phosphate, potassium, magnesium; hypothyroidism; anaemia; malnutrition; metabolic alkalosis reducing drive
- Fluid overload and positive balance: diurese before the next SBT
- Psychological and central: delirium, anxiety, sleep deprivation; excess or reduced respiratory drive

Weaning failure, diaphragm ultrasound, P0.1 and CROP — June Q5 (cont.)

Describe ultrasound assessment of the diaphragm

- Excursion: M-mode, low-frequency probe subcostally through the liver window; normal quiet breathing ≥ 1–1.5 cm
- Excursion < 1 cm suggests dysfunction and higher weaning failure risk; measure during spontaneous breathing (unsupported)
- Thickening fraction: high-frequency probe at the zone of apposition (8th–10th intercostal space), (end-inspiratory − end-expiratory thickness) / end-expiratory × 100
- Thickening fraction < 20–30% predicts SBT failure; low thickness over time indicates ventilator-induced atrophy

Weaning failure, diaphragm ultrasound, P0.1 and CROP — June Q5 (cont.) (cont.)

Explain P0.1 and the CROP index

- P0.1: airway pressure drop in the first 100 ms of an occluded inspiration; reflects central drive, independent of muscle strength
- Normal about 1–2 cmH2O; > 3.5–4 cmH2O suggests high drive/load and likely weaning failure; very low suggests over-assistance
- CROP = [dynamic compliance × maximal inspiratory pressure × (PaO2/PAO2)] / respiratory rate
- CROP ≥ 13 mL/breath/min predicts weaning success; combines mechanics, strength, gas exchange and rate (compare RSBI < 105)

Key point: Diaphragm excursion < 1 cm or TF < 20–30% and P0.1 > 3.5–4 predict failure; CROP ≥ 13 predicts success.

Brain stem death and organ donation — June Q6

List the preconditions and exclusions before brain stem testing

- Known cause of irreversible structural brain damage; apnoeic coma on the ventilator
- Exclude depressant drugs (allow clearance or measure levels), neuromuscular blockade (train-of-four), hypothermia (core temperature > 34 °C)
- Correct metabolic and endocrine causes: Na 115–160 mmol/L, K > 2, glucose 3–20 mmol/L, Mg and PO4 within limits
- Stable circulation (MAP > 60 mmHg), normocapnia and normal pH before testing

Brain stem death and organ donation — June Q6 (cont.)

Describe the clinical brain stem tests and the apnoea test

- Absent pupillary light reflex, corneal reflex, oculovestibular (caloric) reflex
- No motor response in the cranial nerve distribution to central pain; absent gag and cough (tracheal suction)
- Apnoea test: pre-oxygenate with 100% O_2; start $PaCO_2 \geq 6.0$ kPa (45 mmHg), pH < 7.40; disconnect with O_2 insufflation via a catheter or CPAP
- No respiratory effort for 5 min with $PaCO_2$ rising > 0.5 kPa (UK); US criteria $PaCO_2 \geq 60$ mmHg and ≥ 20 mmHg above baseline
- Two senior doctors (≥ 5 years registered, one consultant, not from the transplant team) perform two sets of tests; time of death is after the first set
- Ancillary tests (CT angiography, perfusion scan, EEG) only when the clinical tests cannot be completed

Brain stem death and organ donation — June Q6 (cont.) (cont.)

How would you manage the conflict between the sons about organ donation

- Check for the patient's own wishes first: donor registry, advance directive or what he said; this has priority
- If unknown, ask both sons what their father would have wanted (substituted judgement), not what they want
- Hold a calm family meeting with the specialist donation nurse/team; separate the death conversation from the donation request; allow time
- Follow local law on consent and next of kin; in practice donation does not proceed against a firm objection of close family; document and offer support

Key point: Patient's own wishes first; family conflict → do not proceed, support and document.

Post-ROSC care and temperature control — June Q7

Outline your immediate post-ROSC bundle

- Airway and ventilation: SpO2 94–98% (avoid hyperoxia), normocapnia PaCO2 35–45 mmHg, lung-protective volumes
- Circulation: MAP > 65 mmHg with fluids, noradrenaline ± dobutamine; urine output > 0.5 mL/kg/h, falling lactate; echo
- Coronary angiography: emergent if ST elevation or haemodynamic/electrical instability; otherwise not routine immediately (COACT, TOMAHAWK)
- Glucose control, treat seizures (EEG), sedation, search for the cause (CT head/chest if indicated)

Post-ROSC care and temperature control — June Q7 (cont.)

Compare the TTM1 and TTM2 trials and current guidance on temperature

- TTM1 (2013, n = 950): 33 °C vs 36 °C – no difference in mortality or neurological outcome
- TTM2 (2021, n = 1,900): 33 °C vs normothermia with fever treatment (≤ 37.5 °C) – no difference in 6-month mortality (50% vs 48%) or function; more arrhythmia with hypothermia
- ERC-ESICM: monitor core temperature and actively prevent fever (> 37.7 °C) for at least 72 h in comatose patients
- Hypothermia 32–36 °C is no longer routinely recommended; avoid prehospital cold fluids

How would you manage shivering and hypotension during temperature control

- Assess shivering with the Bedside Shivering Assessment Scale (BSAS)
- Step-up: paracetamol, skin counter-warming, magnesium, buspirone → opioid (pethidine/fentanyl) or dexmedetomidine → propofol → neuromuscular blocker
- Hypotension: cold diuresis and vasodilation on rewarming → fluids guided by responsiveness, noradrenaline
- Myocardial stunning: echo, dobutamine; consider mechanical support in cardiogenic shock

Key point: TTM2: no benefit of 33 °C; prevent fever > 37.7 °C for 72 h.

Neuroprognostication after cardiac arrest — June Q8

When and how should prognostication be done, and how do you avoid bias

- Not before 72 h after ROSC, in a patient with motor score ≤ 3, after excluding sedatives, NMB, hypothermia, hypotension, metabolic causes
- Use a multimodal approach – never a single predictor
- Self-fulfilling prophecy: withdrawal based on a poor prediction makes the prediction come true; use guideline-based criteria and repeat exams
- If uncertain, continue support and reassess; consider organ donation pathway if withdrawal is decided

List the clinical and electrophysiological predictors of poor outcome

- Bilaterally absent pupillary and corneal reflexes at ≥ 72 h (quantitative pupillometry helps)
- Status myoclonus within 72 h (generalised, persistent; EEG to separate from Lance-Adams)
- Bilaterally absent N20 SSEP waves at ≥ 24 h
- Highly malignant EEG after 24 h: suppressed background ± periodic discharges, or burst-suppression

Describe the role of biomarkers and imaging, and the decision rule

- Neuron-specific enolase > 60 µg/L at 48 h and/or 72 h (falsely high with haemolysis)
- Diffuse anoxic injury: CT loss of grey–white differentiation (GWR reduced), MRI diffusion restriction at 2–7 days
- Poor outcome is likely if ≥ 2 of the predictors are present (ERC-ESICM 2021)
- If criteria are not met, outcome is indeterminate: observe and re-evaluate

Key point: ≥ 72 h, multimodal, ≥ 2 predictors; avoid self-fulfilling prophecy.

ICP monitoring and tiered management — June Q9

Discuss indications, monitoring methods and the ICP waveform

- Indications (BTF 4th edition): salvageable severe TBI (GCS 3–8) with abnormal CT; normal CT with ≥ 2 of age > 40, posturing, SBP < 90 mmHg
- External ventricular drain: gold standard, measures and drains CSF; risks infection and haemorrhage
- Intraparenchymal microsensor: easy, lower infection, cannot drain or be recalibrated (drift)
- Waveform P1 (percussion), P2 (tidal, compliance), P3 (dicrotic); P2 > P1 indicates poor compliance; Lundberg A (plateau) waves are pathological
- Non-invasive: optic nerve sheath diameter > 5.7–6 mm, TCD pulsatility index, CT signs – screening only
- Targets: ICP < 22 mmHg, CPP 60–70 mmHg

ICP monitoring and tiered management — June Q9 (cont.)

Describe the tiered strategy to reduce ICP

- Tier 0: head up 30°, neck neutral, sedation/analgesia, normothermia, Na > 135, SpO2 > 94%, PaCO2 35–45, Hb > 7, treat seizures
- Tier 1: CPP 60–70, increase sedation, intermittent CSF drainage, mannitol 0.25–1 g/kg or hypertonic saline boluses
- Hypertonic saline vs mannitol: similar ICP effect; HTS for hypovolaemia/hypotension, mannitol causes diuresis; stop if osmolality > 320 or Na > 155–160
- Tier 2: mild hypocapnia 32–35 mmHg, neuromuscular blockade trial, MAP challenge to test autoregulation
- Tier 3: barbiturate coma (if responsive to test dose), mild hypothermia 35–36 °C, secondary decompressive craniectomy
- RESCUEicp: craniectomy for refractory ICP lowered mortality but increased survival with severe disability
- DECRA: early bifrontal craniectomy for moderate ICP worsened outcome; Eurotherm: early hypothermia harmful
- Avoid in TBI: steroids (CRASH), prophylactic hyperventilation < 30 mmHg, routine hypothermia

Key point: ICP < 22, CPP 60–70; tiered SIBICC; RESCUEicp = less death, more severe disability.

Fluid responsiveness versus fluid tolerance — June Q10

Define fluid responsiveness and fluid tolerance

- Responsiveness: stroke volume or cardiac output rises > 10–15% after fluid; the heart works on the steep part of the Frank–Starling curve
- Only about half of ICU patients are responsive; responsiveness does not mean the patient needs fluid
- Tolerance: ability to receive fluid without organ congestion
- Assess intolerance by venous congestion (VExUS score, hepatic/portal/renal Doppler) and lung ultrasound B-lines, rising EVLW

Fluid responsiveness versus fluid tolerance — June Q10 (cont.)

Explain PPV and the end-expiratory occlusion test in ventilated patients

- PPV = (PPmax − PPmin)/PPmean; > 12–13% predicts responsiveness (grey zone 9–13%)
- PPV limits: needs controlled ventilation without effort, Vt ≥ 8 mL/kg, sinus rhythm, closed chest, normal IAP, HR/RR > 3.6, no RV failure
- EEOT: hold at end-expiration for 15 s raises venous return; CO rise > 5% predicts response
- EEOT works with low Vt and arrhythmia but needs real-time CO monitoring and no spontaneous effort during the hold

Evaluate IVC variation and passive leg raising

- Ventilated: IVC distensibility > 12–18% suggests responsiveness; requires passive full ventilation
- Spontaneous breathing: collapsibility > 40–50% is unreliable (variable effort); unreliable with high IAP, RV failure, tamponade, high PEEP
- PLR: raises legs 45° from semi-recumbent, moves about 300 mL of venous blood; CO/SV rise > 10% predicts response
- PLR is valid in spontaneous breathing and arrhythmia; needs real-time CO measurement (not BP), unreliable with intra-abdominal hypertension

Key point: Responsive ≠ needs fluid; check tolerance (VExUS, B-lines). PLR/EEOT need real-time CO.